IRexp: A database of experimental infrared band lists from open literature

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Abstract

IRexp is a redistributable collection of experimental infrared band lists (cm^{-1} peak positions) mined from open chemistry literature, optionally with $^{1}H/^{13}C$ NMR shifts and resolved structures. The release holds 121,233 records (119,345 PMC Open Access Subset; 1,888 Chemotion/RADAR4Chem), including 43,060 structure-linked entries and 33,201 full IR + ^{1}H + ^{13}C + structure quadruples. IRexp stores numeric band lists, not absorbance traces; each record carries a source DOI and stamped licence / licence-pool fields. Licensing is mixed: Europe PMC joins yield 87,617 commercial (CC-BY/CC0), 20,938 non-commercial (NC*), 10,781 empty/unknown (excluded from the commercial deposit), and 1,897 ShareAlike (Chemotion plus rare PMC SA). The resource is intended for multimodal training, retrieval, and tool-input in spectroscopic agents and autonomous chemistry workflows that need structured, attributable experimental peak lists rather than paywalled PDFs or view-only archives. Complementary elucidation benchmarks are described in a companion research manuscript and are not analysed here.

Keywords: infrared spectroscopy, band lists, PMC Open Access, Chemotion, experimental spectra, data descriptor

1 Background & Summary

Infrared (IR) spectroscopy is routine in organic characterisation, yet open, redistributable collections of *experimental* IR data remain sparse relative to modern machine-learning and agentic tool-use needs. Digitised absorbance libraries such as the NIST Chemistry WebBook [1] and AIST SDBS [2] are valuable but either modest in size or view-only (no bulk redistribution). Computational IR–NMR resources published in *Scientific Data* expand multimodal coverage with simulated spectra [3], while large literature mines for NMR peak lists (notably NMRexp [4]) show that peer-reviewed experimental spectral *lists* with DOI traceability are in scope for this journal. Concurrent literature corpora that include IR among other modalities (for example NMRSpec/NMRTrans [5]) further motivate an IR-focused, redistributable band-list resource rather than another closed spectrum archive.

In 2025–26, spectroscopic AI agents and autonomous chemistry workflows increasingly consume *structured* experimental peak lists as tool inputs and retrieval substrates — not paywalled PDF prose and not view-only web UIs. Agents that plan characterisation, call spectrum tools, or train IR→structure models need redistributable numeric lists with DOI attribution and explicit licence pools (commercial vs non-commercial vs ShareAlike). IRexp is designed as that substrate: training / retrieval / tool-input material for literature-grounded spectroscopic agents, without embedding diagnosis benchmarks or model leaderboards in this Descriptor.

IRexp fills a specific wedge. Experimental sections of chemistry papers conventionally report per-compound IR band lists (wavenumbers in cm^{-1}) together with $^{1}H/^{13}C$ NMR shift lists. That textual convention is the object language models and many elucidation pipelines consume, and it is a different object from a digitised spectrum. IRexp therefore:

1. Harvests open-access full text from the PMC Open Access Subset bulk S3 mirror and peak lists from the Chemotion FT-IR deposit.
2. Extracts deterministic numeric IR band lists (and co-reported NMR strings where present).
3. Resolves compound names to canonical structures with OPSIN [6], RDKit [7], and SELFIES [8] where possible.
4. Releases extracted numbers only — no PDFs, figures, or article full text — with source accessions for attribution.

Among openly redistributable *text-derived IR band lists*, IRexp is large by record count (121,233). It does not claim to replace SDBS or NIST absorbance libraries; SDBS alone holds more structure-linked *spectra* than IRexp holds structure-linked band lists. The intended reuse is multimodal pretraining, supervised IR→structure modelling on `irexp_resolved`, and as the literature substrate for complementary elucidation benchmarks described elsewhere (forthcoming ICLR manuscript on IRSpectra-Bench; cite that work for protocol and model results — not reproduced here) [9].

What this Data Descriptor does not contain.

No hypothesis tests, no large-language-model accuracy tables, and no stage-decomposition of elucidation performance. Those belong in the complementary research paper.

2 Methods

2.1 Source corpora

PMC Open Access Subset.

Primary harvest uses the NCBI PMC OA bulk distribution on Amazon S3 (`s3://pmc-oa-opendata`, HTTPS endpoint `https://pmc-oa-opendata.s3.amazonaws.com`) [10]. Harvest window (recoverable from git snapshots): the bulk S3 IR crawl and `seen_papers` / `ir_harvest_snapshot` artefacts were produced on 2026-06-07 (UTC) (`scripts/s3_ir_harvest.py`; incremental auto-snapshots that day through ∼134,893 raw IR rows). Chemotion was ingested the same day. A harvest snapshot records 188,016 distinct PMC identifiers scanned (`data/irexp/seen_papers.txt.gz`). The released corpus retains records from 15,416 unique `PMC:*` accessions. Extraction operates on open-access plain text objects at flat S3 keys `PMC{id}.{v}/PMC{id}.{v}.txt`. Europe PMC full-text XML is used for some validation re-fetches. The released IRexp DOIs are exclusively `PMC:*` accessions or the Chemotion deposit DOI; no paywalled publisher DOI appears in the 121,233-record file.

Discovery vs `oa_comm` / `oa_noncomm` packages.

Identifier discovery used NCBI E-utilities `esearch` over PMC with an IR-characterisation query and `open access[filter]`, then fetched matching IDs from the flat S3 text layout. The harvest therefore did not pre-filter by walking the PMC OA Subset's `oa_comm` vs `oa_noncomm` vs other package directories. PMC OA is still not a single licence [10]. Licence truth for redistribution is applied post-hoc: every unique PMCID in IRexp (15,416) was joined to Europe PMC `license` metadata (`scripts/join_pmc_licences.py`); each record carries `license` / `license_pool`. Commercial-use (CC-BY/CC0) rows form the Zenodo / Sci Data primary pool; NC* are held aside; empty/unknown are excluded from the commercial deposit — aligning redistributable intent with the OA commercial/non-commercial distinction via article-level licences rather than S3 package paths (`data/NOTICE`; `docs/scientific_data/LICENCE_REMEDIATION.md`).

Chemotion / RADAR4Chem.

1,888 records come from the Chemotion Repository FT-IR collection deposited at RADAR4Chem (DOI `10.22000/OGoEQGlsZGElrgst`) [11], licensed CC-BY-SA-4.0. Ingest (`scripts/chemotion_to_irexp.py`, 2026-06-07): download the MD5-verified deposit; for each of 2,116 ATR-IR analyses, flatten the Quill-delta `content` field to plain text; parse an author-curated IR band list with the same regex extractor and quality gates as the PMC path; resolve the deposit's canonical SMILES with RDKit →

InChIKey + SELFIES; keep the richest band list per InChIKey; drop the 8 molecules already present in the PMC pool → +1,888 new structure-resolved rows. These are not algorithmic peak-picks from absorbance curves; they are author-entered experimental band lists from the ELN deposit, denser than typical paper prose (median 39 bands vs 9 for PMC). Rows carry `license=CC-BY-SA-4.0` and `source_doi` equal to the deposit DOI.

Excluded.

AIST SDBS (view-only; no bulk export) [2]; NIST WebBook join code exists in the repository but contributes 0 records to the released IRexp (`ir_source` is always `experimental` for released rows).

2.2 Discovery and fetch (released corpus)

1. Discover IR-reporting OA PMCIDs via NCBI `esearch` (`open access[filter]` + characterisation-format IR query; year/month sliced).
2. Fetch OA full text from PMC-OA S3 plain-text objects (`PMC{id}.{v}.txt`).
3. Ingest Chemotion deposit author-curated band lists + structures (`scripts/chemotion_to_irexp.py`).
4. Persist harvest provenance in `seen_papers.txt.gz` and optional rawer rows in `ir_harvest_snapshot.jsonl.gz` (134,893 rows including intermediate prose fields; the curated release is `irexp.jsonl.gz`).

Non-OA / Scrapling fence.

Development adapters for ChemRxiv, Beilstein, and generic publisher pages exist for exploration. They are outside the construction path of the released dataset. No released `source_doi` is a paywalled publisher DOI. Methods for IRexp as published here cite only PMC-OA S3 + Chemotion.

2.3 Extraction (band lists, not spectra)

A deterministic regular-expression pipeline (`spectro_scraper/extract.py`) segments experimental text per compound and extracts:

- IR wavenumbers → `ir_bands_cm-1` (list of floats, cm^{-1});
- ^1H and ^{13}C payloads → `h_nmr` / `c_nmr` (author strings, when present).

Gates reject scan-range artefacts and common prose false positives. No curve digitisation is performed: figures are not traced; JCAMP-DX is not stored.

2.4 Structure resolution

Where an IUPAC or systematic name is available, names are converted with OPSIN [6], canonicalised with RDKit [7], and encoded as InChIKey and SELFIES [8]. An optional PubChem [12] name fallback (`USE_PUBCHEM`) handles trivial names. Structure coverage of the full release is 35.5% (43,060 / 121,233). The structure-complete split is shipped as `irexp_resolved`.

2.5 Licence handling

Table 1 Licence pools stamped on the released IRexp corpus.

Pool	Records	Stamp
commercial (CC-BY + CC0)	87,617	`license_pool=commercial`; Zenodo / Sci Data primary
non_commercial (CC-BY-NC*)	20,938	held aside
sharealike (Chemotion + rare PMC SA)	1,897	CC-BY-SA / CC-BY-SA-4.0
empty_unknown	10,781	excluded from commercial deposit

`scripts/join_pmc_licences.py` stamps every row; `scripts/split_license_pools.py` reports provenance and materialises pool files under `data/irexp/licence_pools/`. Narrative and policy: `docs/scientific_data/LICENCE_REMEDIATION.md`.

2.6 Quality tooling

Optional physics gates live in `spectro_scraper/quality.py` (^{13}C peak count ≤ carbon count; ^{1}H integration vs formula; IR wavenumber windows). They were not applied as a hard filter at harvest; Technical Validation reports a full-corpus post-hoc quarantine on `irexp_resolved` (`scripts/quarantine_structure_nmr.py` → `data/audit/structure_nmr_quarantine.jsonl.gz`). Transcription and recall-proxy scripts: `scripts/audit_extraction.py`, `scripts/audit_extraction_recall.py`.

3 Data Records

3.1 Object definition

Each IRexp record is a JSON object. Required chemistry fields for an IR entry:

Not included: absorbance traces, intensities, instrument metadata beyond what appears in source text, PDFs, figures, or full article bodies.

3.2 Files and counts

Paths relative to the project repository / Hugging Face mirror.

Composition of `irexp.jsonl.gz`.

Licence-pool counts (confirmed Track 1 join, 2026-08-26).

Lookup over 15,416 unique PMCIDs; stamped pool files under `data/irexp/licence_pools/`. Sum of pools = 121,233 (unchanged).

Table 2 Core fields of an IRexp JSON record.

Field	Type	Description
`id`	string	Stable record id (InChIKey when resolved, else internal)
`ir_bands_cm-1`	float list	Experimental IR peak positions (cm^{-1})
`ir_source`	string	`experimental` for all released rows
`source_doi`	string	`PMC:<pmcid>` or Chemotion deposit DOI
`h_nmr` / `c_nmr`	string or null	Author-reported shift lists
`smiles` / `selfies` / `inchikey`	string or null	Resolved structure encodings
`has_structure`	bool	Convenience flag
`license` / `license_pool`	string	Per-article stamp (Europe PMC join or Chemotion)

Table 3 Released IRexp files and record counts.

File	Records	Description
`data/irexp/irexp.jsonl.gz`	121,233	Full curated release
`data/irexp_resolved/irexp_resolved.jsonl.gz`	43,060	100% structure-linked
`data/irexp_release/train_no_bench.jsonl.gz`	42,808	Resolved minus IRSpectra-Bench InChIKeys
`data/irexp_release/train_no_bench_nmr.jsonl.gz`	32,949	Same with both ^{1}H and ^{13}C
`data/irexp_release/pretrain_ir.jsonl.gz`	119,345	PMC-only IR pretrain pool
`data/irexp/seen_papers.txt.gz`	188,016 lines	PMC IDs scanned at harvest
`data/irexp/ir_harvest_snapshot.jsonl.gz`	134,893	Intermediate harvest snapshot
`licence_pools/irexp_commercial.jsonl.gz`	87,617	CC-BY + CC0 (Zenodo/Sci Data primary)
`licence_pools/irexp_non_commercial.jsonl.gz`	20,938	NC* held aside
`licence_pools/irexp_sharealike.jsonl.gz`	1,897	Chemotion + rare PMC SA
`licence_pools/irexp_empty_unknown.jsonl.gz`	10,781	Excluded from commercial

The full `irexp.jsonl.gz` remains multi-licence on disk; commercial redistribution must use the commercial pool (or `license_pool == "commercial"`). Hugging Face was re-issued with stamped pools and honest card text (`scripts/publish_hf.py`, 2026-08-26).

Median bands: 9 (PMC), 39 (Chemotion). All 1,360,866 released IR band values fall inside 350–4000 cm^{-1} (full-corpus range check; Technical Validation).

3.3 Access

- Hugging Face: https://huggingface.co/datasets/ilkhamfy/IRexp
- GitHub development mirror: https://github.com/IlkhamFY/spectro-agent

Table 4 Composition of the full curated release.

Slice	Count
All IR band-list records	121,233
With ^{1}H and/or ^{13}C NMR	87,075 (72%)
With resolved structure	43,060 (35.5%)
Structure + any NMR	40,702
Full IR + ^{1}H + ^{13}C + structure	33,201
PMC OA provenance	119,345
Chemotion provenance	1,888
Unique PMC accessions	15,416

Table 5 Licence-pool file counts (Track 1 join).

Pool file	Count	Notes
`irexp_commercial.jsonl.gz`	87,617	CC-BY / CC0 — Zenodo primary
`irexp_non_commercial.jsonl.gz`	20,938	CC-BY-NC* held aside
`irexp_empty_unknown.jsonl.gz`	10,781	empty / unresolved — excluded from commercial deposit
`irexp_other.jsonl.gz`	0	reserved (e.g. CC-BY-ND)
`irexp_sharealike.jsonl.gz`	1,897	Chemotion CC-BY-SA-4.0 (1,888) + rare PMC SA

- Zenodo archival snapshot: DOI pending mint after honest multi-licence metadata.

4 Technical Validation

Machine-readable package: `docs/scientific_data/qc_structure_nmr.json` (and artefacts under `data/audit/`).

4.1 Transcription fidelity (PMC)

On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`), each article was re-fetched from Europe PMC and every recorded wavenumber was checked against the source text ($\pm 1\,\mathrm{cm}^{-1}$ integer match). Result: 560/560 bands and 60/60 records confirmed. Enlarged sample $n = 200$ (same script/seed family, `data/audit/extraction_audit_n200.json`): 2,250/2,261 bands (99.51%) and 196/200 records (98.0%) fully confirmed. This bounds transcription error (hallucinated, mis-parsed, or unit-mangled values).

4.2 Extraction-recall proxy (PMC, harvest path)

A human mark-up of every IR string in every paper remains the gold standard. As an automatic proxy on the actual harvest path (`scripts/audit_extraction_recall.py --n 40 --seed 0`): re-fetch PMC-OA S3 plain text for 40 distinct source

PMCIDs; re-run `extract_records`; compare band-sets to released rows ($\pm 1\,\mathrm{cm}^{-1}$). Result: 2,640/2,640 released bands confirmed in re-extract; 248/248 released IR lists recovered (list-level recall proxy 1.0; 40/40 papers); 4/40 papers yielded *extra* re-extracted lists relative to the curated release (parser over-fire vs post-harvest dedup/curation, not missed released compounds). Summary: `data/audit/extraction_recall_proxy.json`. This is not a substitute for expert human recall.

4.3 Structure–NMR consistency and quarantine (resolved)

Sample (prior).

On 500 `irexp_resolved` records with ^{13}C text (seed 0): 17/500 (3.4%) listed more peaks than carbons. On 500 with ^{1}H text: integrals > formula H+2 in 17/497 (3.4%).

Full resolved corpus.

`scripts/quarantine_structure_nmr.py` applied the same physics gates to all 43,060 structure-linked rows. 1,882 (4.37%) fail ≥1 hard check and are listed in `data/audit/structure_nmr_quarantine.jsonl.gz` (diagnostic only — release files unchanged). Among rows with the relevant modality: ^{13}C peaks > carbons 1,194/34,231 (3.49%); ^{1}H integral > formula+2 1,141/39,672 (2.88%); IR out-of-range 0; unparseable SMILES 0. Re-users should filter quarantined IDs before supervised training. These rates are integrity diagnostics, not an expert skeleton audit.

4.4 IR physical window

Every band in the full 121,233-record release lies in $[350, 4000]\,\mathrm{cm}^{-1}$ (0 / 1,360,866 out of range).

4.5 QC status

Table 6 Technical validation checklist.

Check	Status
Per-PMCID licence join + pool counts	Done — commercial 87,617
Transcription fidelity n=60 and n=200	Done
Extraction-recall automatic proxy (n=40 papers, S3 path)	Done (human recall still optional)
Full-corpus structure–NMR quarantine	Done — 1,882 / 43,060 flagged
Expert structure spot-check ($n \geq 100$)	Optional / deferred

5 Usage Notes

- **Band lists ≠ spectra.** Do not evaluate models trained on IRexp as if they had seen full absorbance curves.
- **AI agents / LLM tool-use.** Prefer the JSON band-list fields and `source_doi` as tool returns or retrieval hits; do not scrape paywalled PDFs or view-only SDBS pages when a redistributable pool already exists. Productized agents should filter by `license_pool` (commercial vs NC* vs ShareAlike) rather than treating the full dump as a single licence.
- **Separate pools by density and licence.** PMC (sparse, author-transcribed) vs Chemotion (denser, author-curated ELN band lists). Keep Chemotion ShareAlike constraints in mind when combining pools; combined redistribution of Chemotion-derived rows must honour CC-BY-SA-4.0.
- **Do not assume PMC = CC-BY-4.0.** Filter to `license_pool == "commercial"` (or use `irexp_commercial.jsonl.gz`) for commercial redistribution; attribute via `source_doi` / `pmcid`.
- **Structure–NMR quarantine.** Before supervised training on `irexp_resolved`, optionally drop IDs in `data/audit/structure_nmr_quarantine.jsonl.gz` (∼4.4% of resolved rows).
- **Training without benchmark leakage.** If using the complementary IRSpectra-Bench problems (companion research paper), fine-tune from `train_no_bench.jsonl.gz` (or rebuild with `scripts/build_train_no_bench.py`) so benchmark InChIKeys are withheld. Protocol and model results live only in that companion manuscript.
- **Structure coverage.** Prefer `irexp_resolved` for supervised structure tasks; 64.5% of records lack SMILES.
- **Attribution.** Cite this Data Descriptor / Zenodo DOI and attribute originating articles through each record's `source_doi`.

6 Data Availability

IRexp numeric extracts are available at:

- Hugging Face Datasets: https://huggingface.co/datasets/ilkhamfy/IRexp
- GitHub: https://github.com/IlkhamFY/spectro-agent (`data/irexp/`, `data/irexp_resolved/`, `data/irexp_release/`)
- Zenodo: archival DOI at proof; primary artifact = commercial pool, `cc-by-4.0` metadata + SA companion

Licensing summary (honest):

- Chemotion (1,888): CC-BY-SA-4.0 [11].
- PMC (119,345): mixed Creative Commons — stamped per article; commercial redistributable 87,617 (CC-BY/CC0); NC* 20,938 held aside; empty/unknown 10,781 excluded from commercial Zenodo (`LICENCE_REMEDIATION.md`).
- Only extracted numeric fields and identifiers are redistributed; source full texts are not.

7 Code Availability

Harvesting, extraction, structure resolution, licence pool splitting, and validation scripts are in https://github.com/IlkhamFY/spectro-agent under the MIT License (`LICENSE`). Principal modules: `spectro_scraper/` (`extract.py`, `normalize.py`, `pipeline.py`, `quality.py`); release harvest `scripts/s3_ir_harvest.py`, `scripts/chemotion_to_irexp.py`; validation `scripts/audit_extraction.py`, `scripts/audit_extraction_recall.py`, `scripts/quarantine_structure_nmr.py`; pool split `scripts/split_license_pools.py`. The version corresponding to this Descriptor will be tagged at Zenodo deposit time.

Author contributions

I.Y.: conceptualization, methodology, software, data curation, validation, writing — original draft.
R.A.V.-H.: conceptualization, supervision, writing — review and editing.

Competing interests

The authors declare no competing interests.

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